

Representative No-Go routes and the evidence that stopped them

Route	Goal	Observed failure	Status
Residual Reuse	Reduce CFG work	Best throughput gain only +1.16%	No-Go
Corrector Gating	Skip physical forwards	~1.5× speed; Stable/NUS and E-hull worsened	No-Go
Budget-aware Gating	Safer corrector skip	Quality gates or speed target failed	No-Go
FN-PRA	Representation alignment	Composition and Stable −6.25 pp	No-Go
CrystalREPA	Unconditional REPA repro	E-hull +0.09424; RMSD worsened	No-Go
RP-QTFG	Training-free physics guidance	Offline direction positive; online RMSD worsened	No-Go
CG-TDR	Teacher residual correction	Residual no better than zero; safe gate near no effect	No-Go
Q1 UQ-PQR	Post-generation quality	Novel −14.86 pp	No-Go
Q2 RFR	Force refinement	Novel −30.25 pp; Unique −8.45 pp	No-Go
Q4 CPRC	Constrained correction	Novel −17.08 pp	No-Go
Q5 CQPS	Quality-preserving selection	Novel −15.63 pp	No-Go
Q6 NS-SetRank	Candidate ranking	Novel −12.50 pp	No-Go
GPU acceleration routes	Bitwise-safe throughput	Batch changed quality; compile/static/MPS below gates	No-Go